

# Simple Harmonic Motion Calculations: A Beginner's Guide

## Lesson 5: Energy in Simple Harmonic Motion

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### Energy in Simple Harmonic Motion - Study Notes

#### Key Concepts

- **Kinetic Energy (KE):** The energy an object possesses due to its motion.
- **Potential Energy (PE):** The energy an object possesses due to its position or condition (in SHM, this is related to the displacement from equilibrium).
- **Total Energy (E):** The sum of kinetic and potential energy in a system undergoing SHM, which remains constant.
- **Equilibrium Position:** The point where the net force on the object is zero, and kinetic energy is at its maximum.
- **Amplitude (A):** The maximum displacement from the equilibrium position, where potential energy is at its maximum.

#### Summary

Simple Harmonic Motion (SHM) is characterized by a continuous exchange of energy between kinetic and potential forms. As an object oscillates, it constantly converts potential energy into kinetic energy and vice versa. The total energy of the system remains constant throughout the motion, meaning energy isn't lost (assuming no friction or damping forces). At the equilibrium position, the object moves fastest, resulting in maximum kinetic energy and zero potential energy. Conversely, at the points of maximum displacement (amplitude), the object momentarily stops, possessing maximum potential energy and zero kinetic energy.

The total energy in SHM can be calculated in two ways: using the mass, velocity, and spring constant ( $E = (1/2)mv^2 + (1/2)kx^2$ ) or using the spring constant and amplitude ( $E = (1/2)kA^2$ ). The latter is often more convenient as it only requires knowing the spring constant and amplitude, which are typically constant values for a given system. Understanding this energy exchange is fundamental to analyzing and predicting the behavior of systems undergoing SHM.

#### Important Points

- **$KE = (1/2)mv^2$ :** Kinetic energy depends on the mass ( $m$ ) and the square of the velocity ( $v$ ).
- **$PE = (1/2)kx^2$ :** Potential energy depends on the spring constant ( $k$ ) and the square of the displacement ( $x$ ) from equilibrium.
- **$E = (1/2)mv^2 + (1/2)kx^2 = \text{constant}$ :** Total energy is the sum of KE and PE and remains constant in ideal SHM.
- **Maximum KE:** Occurs at the equilibrium position ( $x=0$ ).
- **Maximum PE:** Occurs at the maximum displacement ( $x = \pm A$ ).
- **$E = (1/2)kA^2$ :** A useful alternative formula for total energy, especially when velocity is unknown.
- **Conservation of Energy:** In ideal SHM (no friction), energy is conserved; it's just transformed between KE and PE.

## Examples

### **\*\*Example 1: Spring-Mass System - Finding Total Energy\*\***

A spring with a spring constant ( $k$ ) of 100 N/m has a mass ( $m$ ) of 0.5 kg attached. What is the total energy of the system?

- **\*\*Solution:\*\*** We can use the formula  $E = (1/2)kA^2$ . We need to know the amplitude ( $A$ ). Let's assume the amplitude is 0.1 meters ( $A = 0.1$  m).

- \*  $E = (1/2) * 100 \text{ N/m} * (0.1 \text{ m})^2$

- \*  $E = (1/2) * 100 * 0.01$

- \*  $E = 0.5 \text{ Joules}$

### **\*\*Example 2: Analyzing Energy at Different Points\*\***

Consider the same spring-mass system ( $k=100$  N/m,  $m=0.5$  kg,  $A=0.1$  m). Describe the energy distribution at the equilibrium position and at the maximum displacement.

- **\*\*Solution:\*\***

- \* **\*\*Equilibrium Position ( $x=0$ ):\*\*** At this point, the displacement is zero, so  $PE = 0$ . All the energy is in the form of KE. Therefore,  $E = KE$ .

- \* **\*\*Maximum Displacement ( $x = \pm 0.1$  m):\*\*** At this point, the displacement is equal to the amplitude, so PE is maximum.  $KE = 0$  because the mass momentarily stops. Therefore,  $E = PE$ .

## Quick Review

1. What is the formula for kinetic energy?
2. What is the formula for potential energy in SHM?
3. Why is the total energy in SHM considered constant (in an ideal system)?
4. Where is the kinetic energy at its maximum in SHM?
5. If a spring has a spring constant of 200 N/m and an amplitude of 0.2 m, what is the total energy of the system? ( $E = (1/2)kA^2$ )

## Further Reading

- **\*\*Damped Harmonic Motion:\*\*** Explore what happens when energy is lost from the system (e.g., due to friction).
- **\*\*Driven Harmonic Motion:\*\*** Investigate what happens when an external force is applied to the system.
- **\*\*Pendulum Motion:\*\*** Learn how the principles of SHM apply to the motion of a simple pendulum.
- **\*\*Resonance:\*\*** Understand the phenomenon of resonance and its implications in SHM systems.

